PATH ANALYSIS AND STRUCTURAL EQUATION MODELING

Part I

Basic Statistical Modeling Context

			Outcome/Dependent Variable	
			Continuous	**Categorical**
Predictor Variables	Observed	**Continuous**	Regression	Logistic
		Including moderation & mediation		
		Categorical	MANOVA/Regression	Non-parametric (e.g. chi-square)
	Latent	**Continuous**	Factor Analysis	Item Response Theory
		Categorical	Latent Profile Analysis	Latent Class Analysis

Finite Mixture Models

INTRO TO SEM

- SEM is AKA analysis of covariance structure, covariance structure modeling, and causal modeling
  - **Note**: the design of the study allows for causal inference, not the SEM framework per se
- SEM is a growing family of procedures not a single procedure
- These methods fall under the umbrella of SEM
  - Path analysis, **CFA**, multilevel SEM, and latent variable modeling (**latent class analysis**, latent transition analysis, latent growth modeling)

KEY CONCEPTS

- Structural equation modeling is a collection of regression-based statistical procedures that allow the examination of relationships of observed as well as ***unobserved variables***
  - **Path analysis (previously)** can be considered a special case of SEM with **only observed variables**
  - **(Confirmatory/Exploratory) Factor analysis** can be considered a special case of SEM with **measurement models to model relationships between observed measures and latent variables, also called factors**
  - **Latent Class analysis aims to do the same as FA however the variables are measured categorically**
- The goal was to reduce a large number of variables into identifiable clusters of variables to better understand the structure of the data

TERMINOLOGY

- Exogenous variable (similar to independent variable)
- Endogenous variable (similar to dependent variable)
- Fixed (constrained) or Free Parameters
- Goodness-of-Fit
- Latent variable
- Measurement error: the difference between a measured quantity and its true value
  - Recall two types of measurement error (?)
  - True score theory assumes that observations are comprised of their true value and some random error

SOME NOTATION

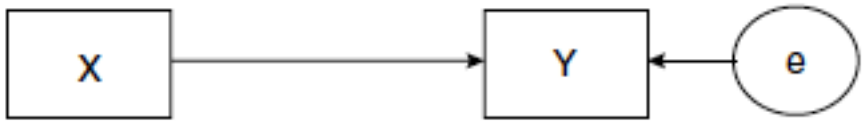
- Observed Variables are denoted by a box $\boxed{\text{V}}$
- Latent Variables or Factors are denoted by a circle F
- **E**rror variance in the observed Vs
- Disturbances in LVs (unexplained variances – variance not explained by the model)

$\longrightarrow$ Directional: Hypothesized structural/causal effect

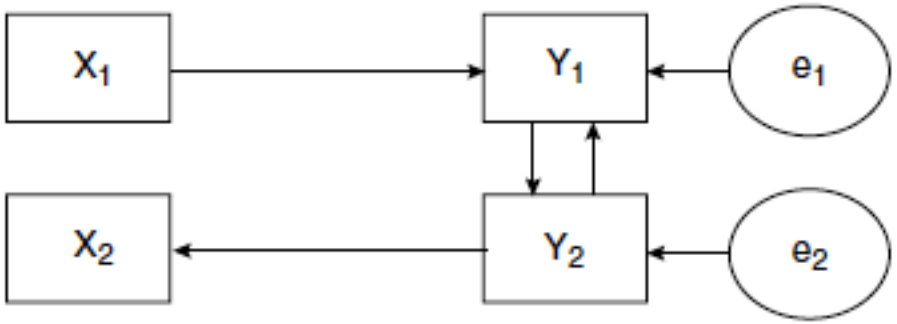
$\longleftrightarrow$ Non-Directional: Hypothesized non-structural/non-causal variation

CHARACTERISTICS: PATH ANALYSIS

- Path analysis falls under the umbrella of SEM and allows researchers to test models that include only observed variables
- The most basic type of path analysis model is one where an exogenous variable (*X*) has a direct influence on an endogenous variable (*Y*)

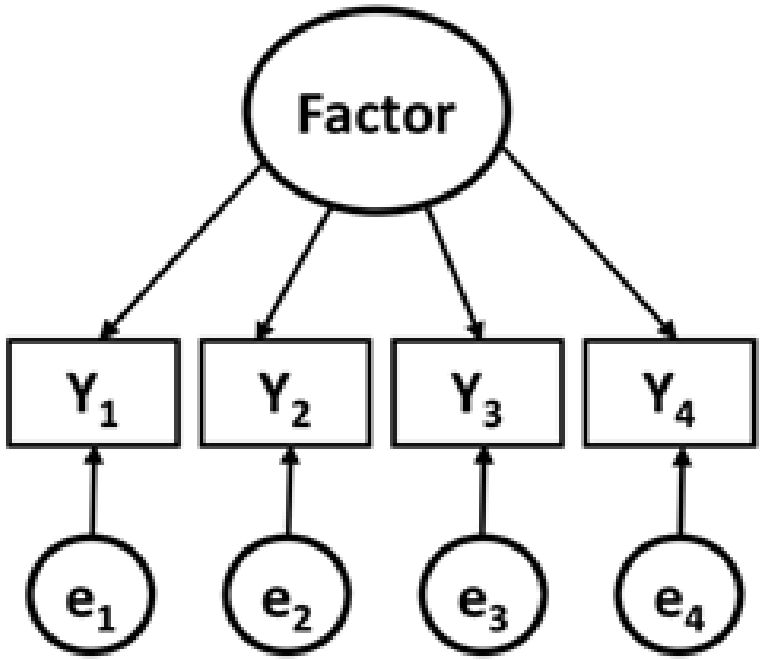


Recursive model: NO feedback loops



Non-Recursive model

EXPLORATORY FACTOR ANALYSIS: THE BEGINNING



In FA some underlying factors are responsible for producing the observed variables with error

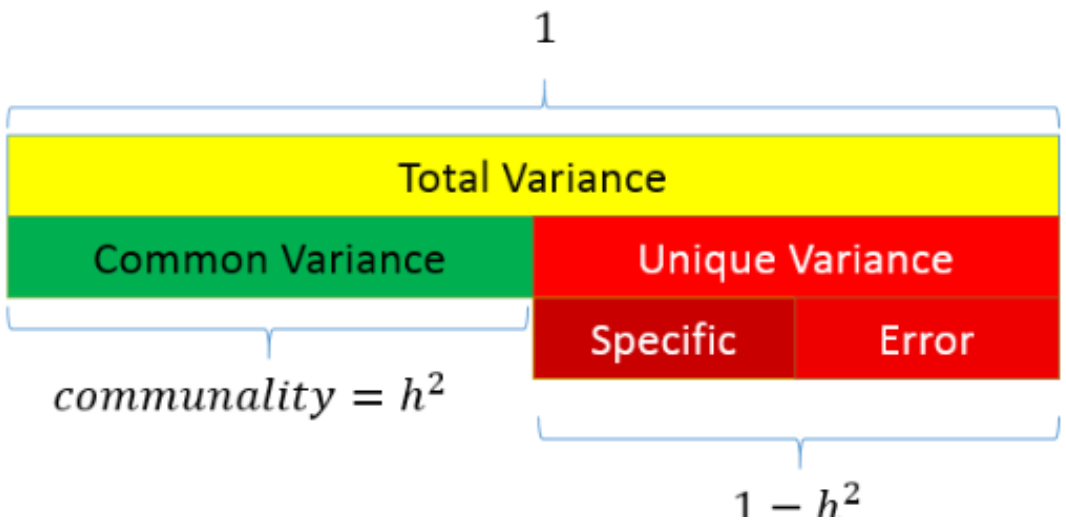
EFA aims to reproduce the observed relationships among a set of observed indicators with a smaller set of latent factors or variables

The factor is the reason for the covariation between the observed variables

<u>The RQ: How many factors best characterize the data</u>

Notice the direction of the arrows

FACTOR ANALYSIS



- Factor analysis partitions the variance of each indicator into two parts
  - ***Common variance***: variance accounted for by the latent variable(s) which is based on the variance shared with other indicators; and
  - ***Unique variance:*** variance specific to the indicator + measurement error

EXPLORATORY FACTOR ANALYSIS

- The intention of factor analysis is to determine the number and nature of the latent variables (also referred to as ‘factors’) that account for the variation and covariation among observed variables (aks ‘indicators’ or ‘manifest variables’)
  - This is exploratory but data driven
- A latent variable is an unobserved variable which influences more than one observed measure and accounts for the correlations among the observed variables

COMMONALITY OF EFA & LCA

- Both provide more parsimonious ways to view relationships among indicators as the number of latent variables is less than the number of observed variables

SEM PROCEDURES

- Model specification
  - In path analysis and other SEM procedures, the researcher specifies relationships between variables a priori given existing theory and previous empirical research
- Model identification
  - Complicated: boils down to how large your sample is and how many parameters are being estimated
  - Depends on the degrees of freedom (see next slide)
- Model estimation and evaluation
  - Estimating the parameters of the model and determining model fit
- Model modification
  - If the model has a poor fit, determine the problem and re-estimate
  - A highly iterative and time consuming process

WHAT ARE ‘DEGREES OF FREEDOM’

- **Defined:** the number of parameters that can vary ‘freely’

	At risk	Not at risk	
Exposed	6	10	16
Not exposed	4	10	14
	10	20	30

MODEL SPECIFICATION

- An exploration of how variables are interrelated
- Factor loadings
  - A factor loading reflects the relationship between a latent construct and an observed variable
  - ***EFA***: reflected by the slope of increase (when positive) or decrease (when negative) for each unit of increase in the latent construct (i.e., a regression of the latent variable ON the observed variable)
  - ***LCA***: reflected by the item response probability, the probability of belonging to the latent factor given the set of responses on the observed variable
- Unique variance is variance that is not accounted for by the latent construct and is a combination of systematic variance specific to the observed variable and random error variance (i.e., measurement error or unreliability in the observed variable)

MODEL EVALUATION AND INTERPRETATION: GOODNESS-OF-FIT INDICES AND PARAMETER ESTIMATES

- Evaluating model fit involves examination and interpretation of both overall model fit as well as parameter estimate evaluation
- There are a large number of goodness-of-fit statistics

MODEL TEST STATISTICS

- Model test statistics are tests of inference related to the fit of the SEM model
- Chi-square or likelihood ratio chi-square test tests the null hypothesis that the predicted covariance matrix and the sample variance-covariance matrix are equal
  - You want to not reject this, but almost always will
  - This is most useful when comparing nested models

APPROXIMATE FIT INDICES

- Comparative measures of fit include AIC, BIC, aBIC
- Absolute fit indices include the SRMR, RMR, and RMSEA
- Incremental fit indices evaluate model fit to a baseline model (CFI, TLI, NNFI, NFI).

MULTIPLE GROUP MODELS

- Tests for differences in parameter estimates across groups within a latent variable framework.
- The theoretical model is established with the data first, and then separate models are generated by group.
- Nonstatistically significant chi-square tests for each group indicate similarity of the groups

Latent Class Analysis

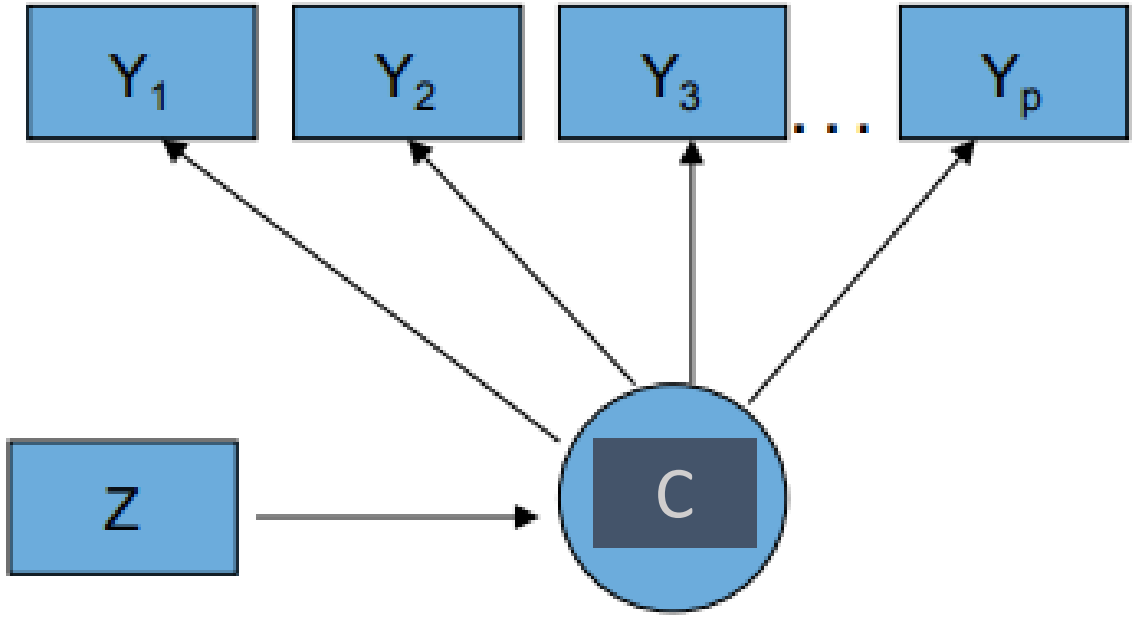
SEM for categorical variables

When to use Latent Variable Models

- Cross-sectional data
  - Creating typologies of respondents
  - Example: adolescent sexual-risk taking behavior (?)
- Longitudinal data (developmental studies)
  - Growth mixture models
    - Typologies of “growth”: how does depression manifest over the life course, not one size fits all
  - Latent transition (“Hidden Markov”) models
    - Do youth go ‘in and out’ of bullying “states” with the bullying defined as a latent construct
- Clustered data
  - Multilevel latent class model
    - Typologies of bullies nested within schools
  - Multilevel latent transition model
    - Transitions of bullies nested within schools
- JASP/JAMOVI and R will conduct all of these!

The Latent Class Model

- Observed Categorical Items (*Y*'s)
- Categorical Latent Class Variable (*C*)
- Continuous or Categorical Covariates (*Z*)



Person centered v. Variable centered analysis

- LCA is a person-centered approach
- Variable centered (e.g., EFA)
  - Correlations/covariances are of interest
  - Structure among columns (e.g., factor)
- Person centered
  - Structure among rows is of interest
  - Seeks to group individuals based on shared characteristics
  - Goal: identify qualitatively distinct groups

Factor 1 Factor 2

	child_id	ScaleA	ScaleB	ScaleC	ScaleD	ScaleE
Group 1	101	4.00	4.57	1.75	4.50	4.40
	102	4.88	4.57	5.00	4.88	4.40
	103	3.63	4.14	2.25	4.13	3.80
	104	4.00	3.57	2.00	3.25	.60
	105	3.75	4.14	2.50	3.88	3.40
	106	4.63	4.57	2.50	5.00	4.60
	107	4.63	4.29	3.75	4.13	3.80
	108	4.25	3.71	3.50	4.75	4.20
Group 2	109	4.88	4.71	4.75	4.50	4.80
	110	4.50	4.71	3.75	4.75	5.00
	111	3.63	4.00	4.50	2.63	3.80
	112	3.88	4.00	4.00	3.75	3.60
	113	5.00	5.00	1.50	5.00	5.00
Group 3	114	3.88	5.00	2.50	3.75	3.40
	115	4.00	4.57	3.75	4.75	3.80
	116	2.75	3.57	.50	3.00	2.00
	117	5.00	5.00	4.25	5.00	4.40
	118	4.13	2.57	1.00	3.00	4.00
	119	.86	4.00	1.25	.00	2.33
	120	3.25	4.29	4.25	1.71	2.80
	121	3.75	3.29	.75	4.00	4.40

Mixture Models

- LCA is a special case of mixture model
- Modeling a “mixture” of sub-groups within a population
- Assumes
  - “Finite” number of homogeneous categories
  - The population is a mixture of qualitatively different groups of individuals
  - Groups can be identified based on similarities in response patterns
- RQ: Can we identify subgroups of individuals who fit various ‘profiles’

Quick examples

- **Latent**: substantively or by degree of syndrome
- **Example**: different types of victimization, psychological and psychosocial risk factors are often co-occurring
- The **Frailty construct** in aging research
  - **Manifest**: criteria (as derived from the literature)
    - Weight loss above threshold
    - Low energy expenditure
    - Low walking speed
    - Weakness beyond threshold
    - Exhaustion
  - RQ?

Aside (substantively or by degree?)

	Class 1	Class 2	Class 3
Weight Loss	90%	10%	0
Weakness	10%	95%	0
Slowness	10%	0	95%
Low Physical Activity	95%	10%	0
Exhaustion	.1%	95%	0

	Class 1	Class 2	Class 3
Weight Loss	95%	50%	0
Weakness	95%	50%	0
Slowness	95%	50%	0
Low Physical Activity	95%	50%	0
Exhaustion	95%	50%	0

Latent class analysis

- Seeks homogeneous subpopulations, *how*?
- Features that characterize latent groups
  - Prevalence in overall population
  - Proportion endorsing each item (i.e., because the data are categorical)
    - If LPA → mean of item
  - Number of items
    - Fewest number to achieve homogeneity / conditional independence

What is frailty among older adults?

How would you interpret and label these classes?

Criterion/Item	3-Class Model		
	Class 1	Class 2	Class 3
Weight Loss	.072	.11	.54
Weakness	.029	.26	.77
Slowness	.004	.45	.85
Low Physical Activity	.000	.28	.70
Exhaustion	.027	.16	.56
Class prevalence (%)	39.2	53.6	7.2

Bandeen-Roche et al., J Gerontol, 2006

Motivating Example: Patterns & Classes

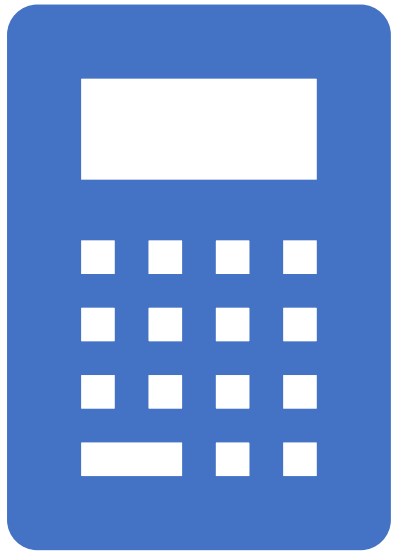
Y1: CRT	Y2: CGT	Y3: CCT	Observed Frequency	Observed proportion n/N
Yes	Yes	Yes	696	.406
Yes	Yes	No	68	.040
Yes	No	Yes	275	.161
Yes	No	No	130	.076
No	Yes	Yes	34	.020
No	Yes	No	19	.011
No	No	Yes	125	.073
No	No	No	366	.214

1-C Model

Y1: CRT should be taught in schools? (1=Y/2=N)
Y2: CGT should be taught in schools? (Y/N)
Y3: CCT should be taught in schools? (Y/N)

2-C model

	Class 1	Class 2
CRT (Y1)		
Yes P(Y1=1|C)	0.96	0.23
No P(Y1=2|C)	0.04	0.77
CGT (Y2)		
Yes P(Y2=1|C)	0.74	0.04
No P(Y2=2|C)	0.26	0.96
CCT (Y3)		
Yes P(Y3=1|C)	0.92	0.24
No P(Y3=2|C)	0.08	0.76
Class prop.	0.6205	0.3795



Model equation for 2-class LC model for 3 indicators

- Model for *Y*'s = 1 (Yes) and 2 (No)

The probability of a particular response pattern

Example

(Yes, Yes, Yes)

(Yes, No, Yes)

…

Etc

$$\Pr(y_1 = Y_1, y_2 = Y_2, y_3 = Y_3)$$

Two key model assumptions

1. Mixture assumption
   - There is a latent categorical variable (<u>NOT MANIFEST VARIABLE</u>) that consists of a finite number of types of people, i.e. classes
2. Local independence assumption
   - Within classes responses are ***conditionally*** independent
   - No relationship between observed variables once the latent class is present, i.e., the latent class is the REASON for the interrelationship between variables (may or may not be realistic)

Example

- Think of what constitutes (constituents of) PTSD?
  - Mixture assumption: we can identify subgroups of persons with different levels of PTSD indicators such as startle response, avoidance behaviors, etc. This implies that PTSD is NOT a unidimensional concept
  - Local independence assumption: the *class specific* relationships between PTSD indicators are independent once we identify the class
    - Let's say we identify a class of people with high scores on avoidance of people' and 'avoidance of places'– this assumption means that are highly likely to avoid people and places these two variables are no longer related – the underlying latent construct is the cause of these indicators – we might label it as 'Avoidance'…
- Neither of these may be reasonable assumptions but they can be tested by our models.

Choose correct analysis for the RQ

- There is general agreement that PTSD symptoms should reflect their underlying common cause (i.e., PTSD) (McNally et al., 2015, Cloitre et al., 2014). The transdiagnostic approach is one organizing framework that focuses on identifying those psychological processes that underlie, or cause, PTSD (Gutner et al., 2016). To do so, latent variable models are commonly used to explore the relationship between the manifest indicators of PTS and the underlying latent variable PTSD. However, "there are major limitations to these latent variable approaches to psychopathology (McNally et al., 2015, p. 838)." First, scholars have cast doubt about whether one single factor can explain the heterogeneity of mental disorders; rather the etiology of mental illness is believed to be complex and multifactorial (Robinaugh, et al., 2014; Armour, Fried, Deserno, Tsai, et al., 2017; McNally, Heeren, & Robinaugh (2017); Sullivan, et al., 2018). Second, latent variable models assume local independence between symptoms, meaning that any covariance between symptoms is accounted for by the underlying latent variable (i.e., PTSD). This assumption has been deemed implausible (McNally et al., 2015) since it is unlikely that PTS symptoms are independent of each other after controlling for PTSD. Rather, it is more likely that PTS symptoms remain highly interconnected.

Barboza, G. E., & Valentine, R. (2022). A network analysis of post-traumatic stress among youth aging out of the foster care system. *Children and Youth Services Review*, *140*, 106589.

How is sexual risk-taking behavior different?

Latent Class Analysis				
Class Prev.	0.177	0.120	0.589	0.114
	Class 1: Normative	Class 2: Oral Normative	Class 3: Abstainers	Class 4: Early Timing; Intoxicated first time; multi-partner
Oral intercourse				
No oral intercourse	0.037	0.111	0.095	0
Normative	0.896	0.575	0.856	0.617
Early	0.066	0.313	0.048	0.382
Timing First Sex				
Never had sex	0	0.392	1	0
Normative	0.984	0.252	0.003	0.421
Early	0.015	0.355	0	0.578
Intoxicated First Time				
Never had sex	0.554	0.774	0.988	0.064
Not intoxicated	0.242	0.144	0.006	0.329
Was intoxicated	0.202	0.081	0.005	0.606
Condom Use				
Never had sex	0.818	0.975	0.998	0.766
Used Condom	0.093	0	0.001	0.161
Did not use	0.088	0.025	0.0000	0.072
Sex Partners				
None	0.746	0.603	0.986	0.473
One	0.164	0.193	0.013	0.198
More than one	0.089	0.204	0	0.328

Example: model-implied proportion

	C=1	C=2
P(C)	.620	.380
P(Y1=1|C)	.960	.229
P(Y2=1|C)	.742	.044
P(Y3=1|C)	.917	.240

Y1: CRT	Y2: CGT	Y3: CCT	Observed Frequency	Observed proportion n/N
Yes	Yes	Yes	696	.406
Yes	Yes	No	68	.040
Yes	No	Yes	275	.161
Yes	No	No	130	.076
No	Yes	Yes	34	.020
No	Yes	No	19	.011
No	No	Yes	125	.073
No	No	No	366	.214

P(Y1=1, Y2=1, Y3=2) =
P(Y1=1, Y2=1, Y3=2 | C=1) .620 +
P(Y1=1, Y2=1, Y3=2 | C=2) .380

(0.960) (0.742) (1-0.917) (0.620) +
(0.229) (0.044) (1-0.240) (0.380) ≈ .04

Implied is very close to observed

Evaluating Model Fit/Selecting the # of classes

- a priori theory
- Chi-Square goodness of fit
- Classification error
- Information Statistics – AIC, BIC, others
- Lo-Mendell-Rubin (LMR) – (normal *Y*)
- Bootstrapped Likelihood Ratio Test

Entropy

- Similar to our classification ‘error’ table from chi-square and logistic regression
- Measures classification error
- Ranges between 0 – 1 (1 = perfect classification)

Information Criterion Statistics

- AIC: -2LL + 2s
- BIC = -2LL + s*log(N)

where s = # of parameters, N = sample size

Here, as in **all SEM models**, smaller values are better

BIC is typically preferred

In LVM, the BIC statistic is the gold standard

1. Evaluate Model Fit

GLOBAL FIT

LOCAL FIT

SUBSTANTIVE CRITERIA

1a. Global fit

- H0: model with C classes; H1: saturated model
- We use **log likelihood and chi-square statistic**
- df = number of patterns -1 - Npar

1a. Information Criteria

- for model comparison
- parsimony versus fit
- Common criteria
  - BIC(LL) = -2LL + ln(N) * Npar
  - AIC(LL) = -2LL + 2 * Npar

Model fit Comparisons

	L2	BIC	AIC	df	P-value
1-C					
2-C					
3-C					
4-C					
5-C					
6-C					

Note: Can make elbow plot of AIC/BIC etc to show lowest value

Example

Political Participation in Greek Elections

Political Participation

- There are different ways of trying to improve things in [Greece] or help prevent things from going wrong. During the last 12 months, have you done any of the following? **(1 = Yes, 2 = No)**

Greece • documents and data files

ESS Round 5 - 2010

Data files

Country file (subset of integrated file)

Parents' occupation - country file

Sample data (SDDF)

Documents

ESS5 Questionnaires GR

ESS5 Contact form GR

ESS5 Showcards GR

ESS5 Advance letters GR

ESS Round 4 - 2008

Data files

Country file (subset of integrated file)

Parents' occupation - country file

Documents

ESS4 Main and Interviewer Questionnaire GR

ESS4 Supplementary Questionnaires GR

ESS4 Contact Form GR

ESS4 Showcards GR

ESS4 Fieldwork Instructions GR

ESS4 Advance Letters GR

ESS4 Press Release GR

62	contplt	Numeric	1	0	Contacted politician or government official last 12 months	{1, Yes}...	6 - 9	9
63	wrkprty	Numeric	1	0	Worked in political party or action group last 12 months	{1, Yes}...	6 - 9	9
64	wrkorg	Numeric	1	0	Worked in another organisation or association last 12 months	{1, Yes}...	6 - 9	8
65	badge	Numeric	1	0	Worn or displayed campaign badge/sticker last 12 months	{1, Yes}...	6 - 9	7
66	sgnptit	Numeric	1	0	Signed petition last 12 months	{1, Yes}...	6 - 9	9
67	pbldmn	Numeric	1	0	Taken part in lawful public demonstration last 12 months	{1, Yes}...	6 - 9	8
68	bctprd	Numeric	1	0	Boycotted certain products last 12 months	{1, Yes}...	6 - 9	8

How many patterns are possible with 7 observed variables?

2	2	2	2	2	2	2
_____	_____	_____	_____	_____	_____	_____

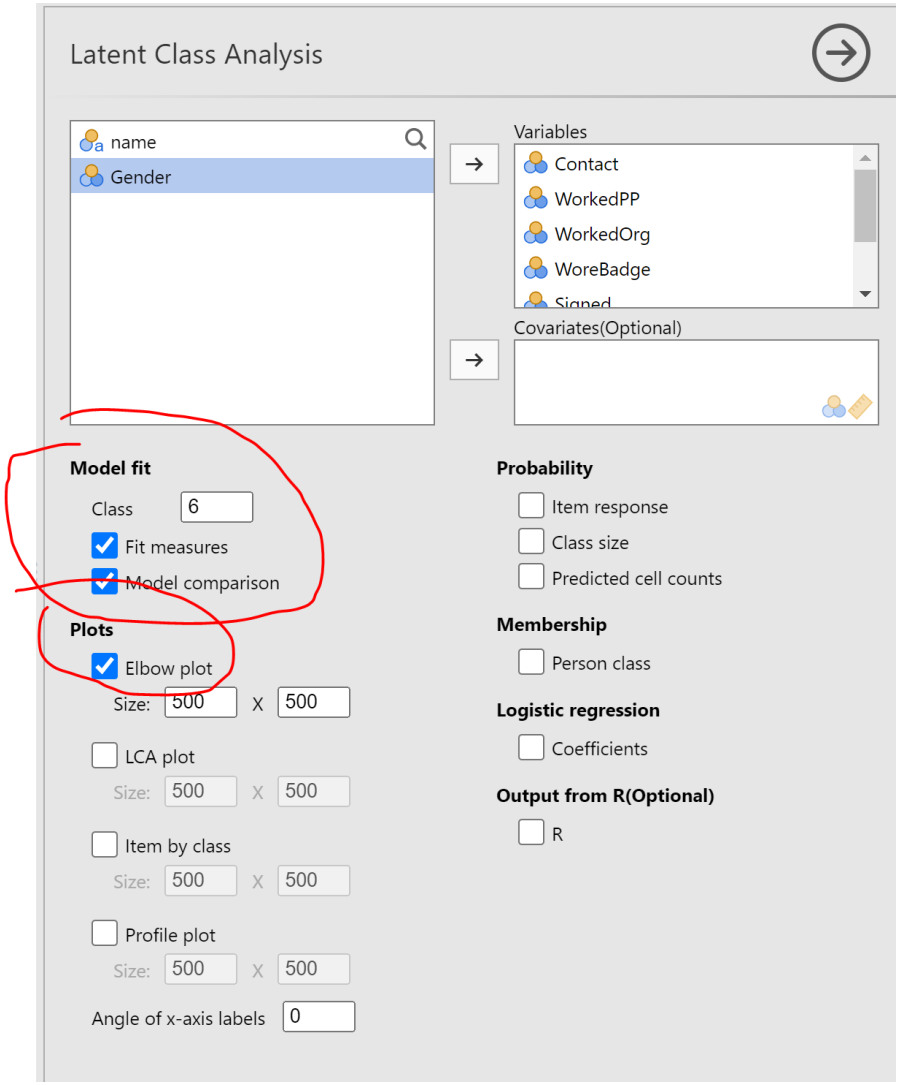
$2^7 = 128$ possible response patterns

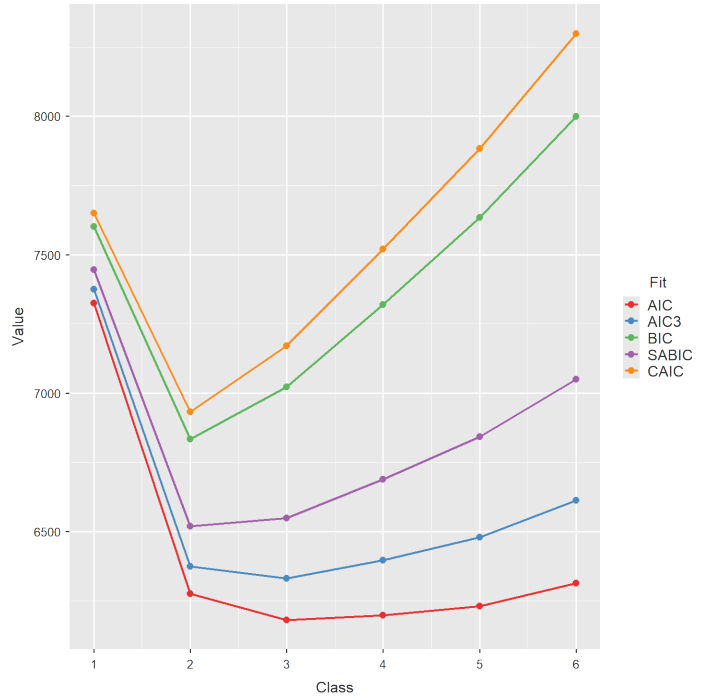
Recall from last week → more unique parameters than patterns → model tested
i.e. the model has many fewer parameters than patterns

Political Participation in Greek Elections

- Data from the European Social Survey is located https://ess.sikt.no/en/?tab=overview
- Let's estimate the model
- We will estimate between 1 and 5 classes and create a typology of Greek political participation
    - Note: this always is done first
    - DO THIS IN JAMOVI!!! Here's why…
        - To do so, install 'snowLatent' and 'snowRMM'

snowRMM→Latent Class Analysis



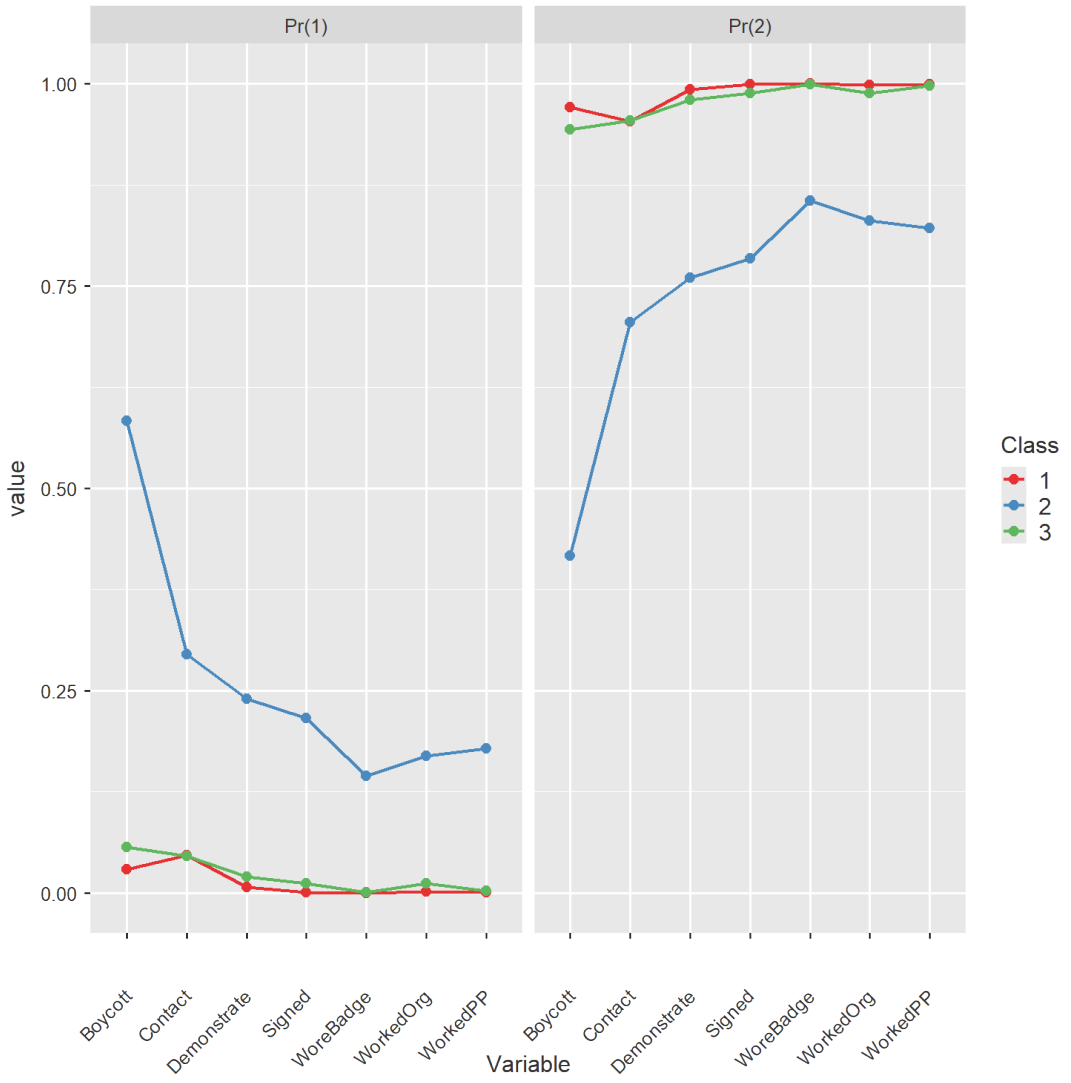


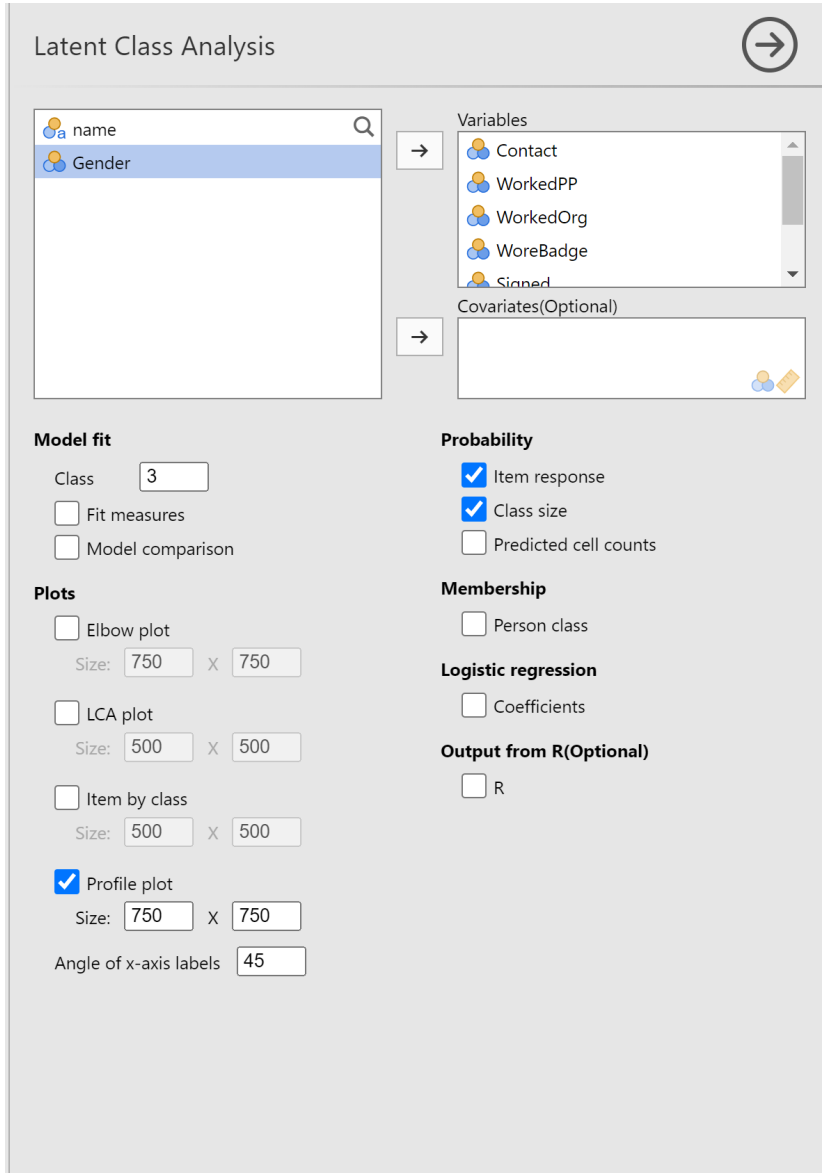
Model fit Comparisons

- Note: Can make elbow plot of AIC/BIC etc to show lowest value

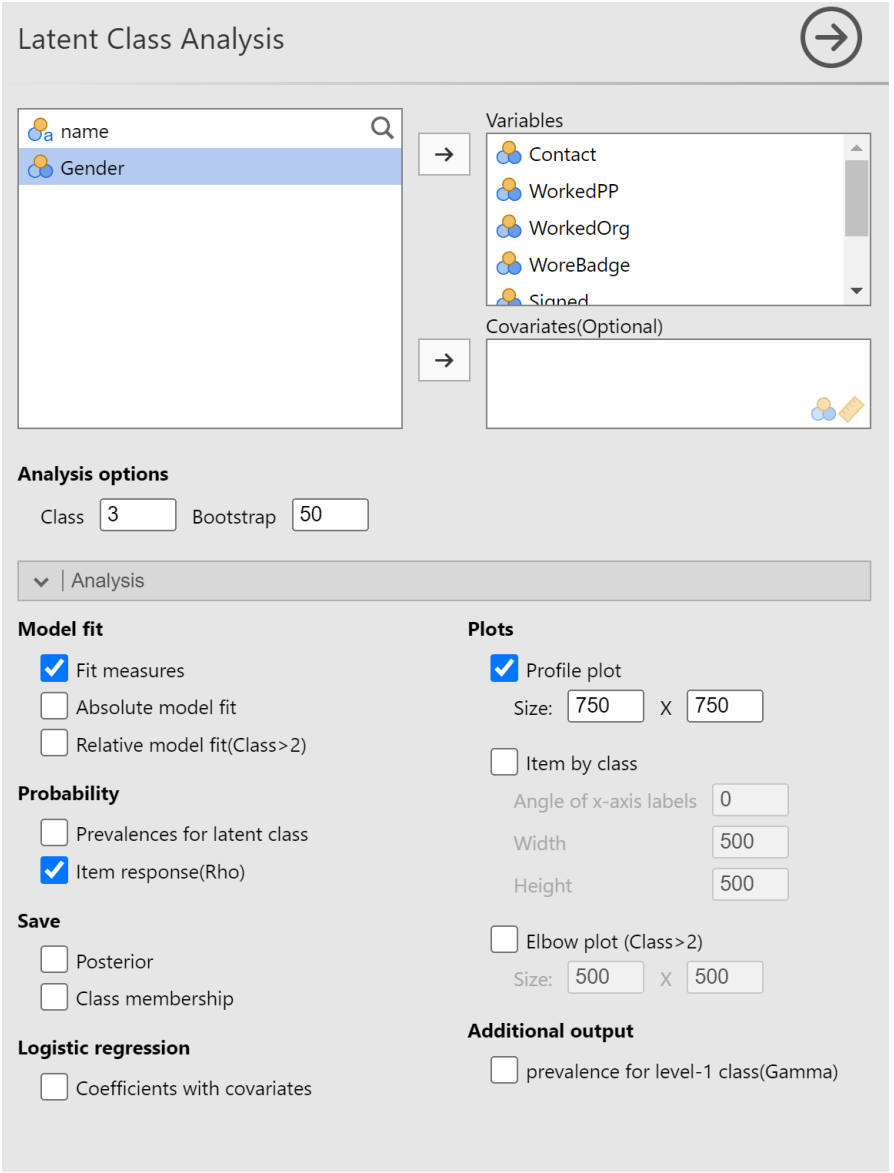
Model comparison								
Class	AIC	AIC3	BIC	SABIC	CAIC	Log-likelihood	χ^2	G^2
1	6774.36	6781.36	6813.82	6791.58	6820.82	-3380.18	15808663.68	1281.62
2	5767.44	5782.44	5851.98	5804.32	5866.98	-2868.72	358.26	269.00
3	5731.71	5754.71	5861.34	5788.27	5884.34	-2842.85	244.02	217.33
4	5736.34	5767.34	5911.07	5812.58	5942.07	-2837.17	287.47	206.81
5	5726.95	5765.95	5946.77	5822.86	5985.77	-2824.48	176.87	181.44
6	5732.82	5779.82	5997.73	5848.40	6044.73	-2819.41	153.53	171.16
You want to select the model with the lowest fit statistic ideally								

3. Interpreting the results and using substantive criteria



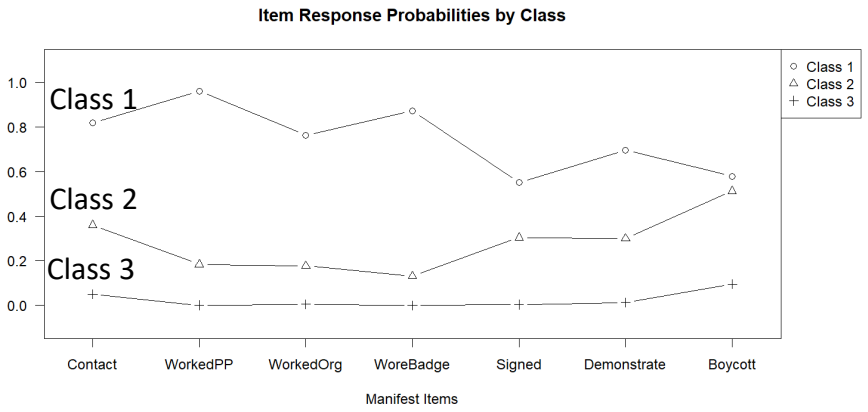


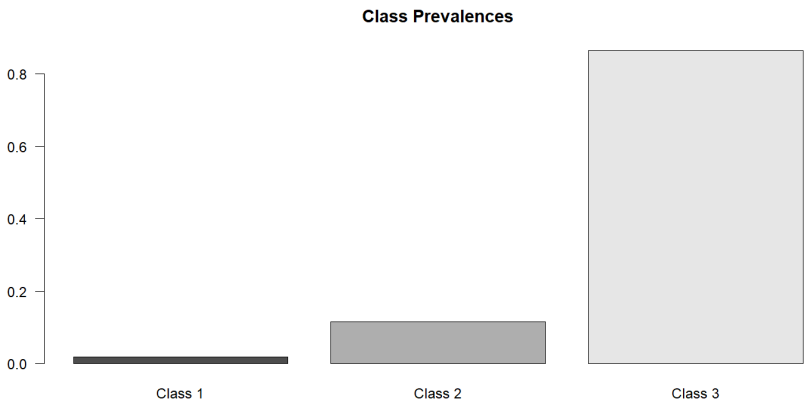
Use snowLatent



Probability of **Contact**

	Y = 1	Y = 2
Class 1	0.82	0.18
Class 2	0.36	0.64
Class 3	0.05	0.95





Model fit evaluation: summary

- Compare models with different number of classes using BIC, AIC, bootstrapped L2
- Evaluate overall fit using bootstrapped L2 and bivariate residuals
- Can be useful to look at the profile of the different solutions: if nothing much changes, or very small classes result, fit may not be as useful

Prediction

- We are interested in how the covariate predicts membership in a latent class

Adding covariates

- Now we can include covariates to predict membership in each latent class
- Note: you must choose the latent classes first, before including covariates!
- We are predicting membership in a latent class, 1 = in the class, 0 = not in the class → logistic regression
  - With more than two classes, we have more than two categories to predict
  - So, we use logistic regression

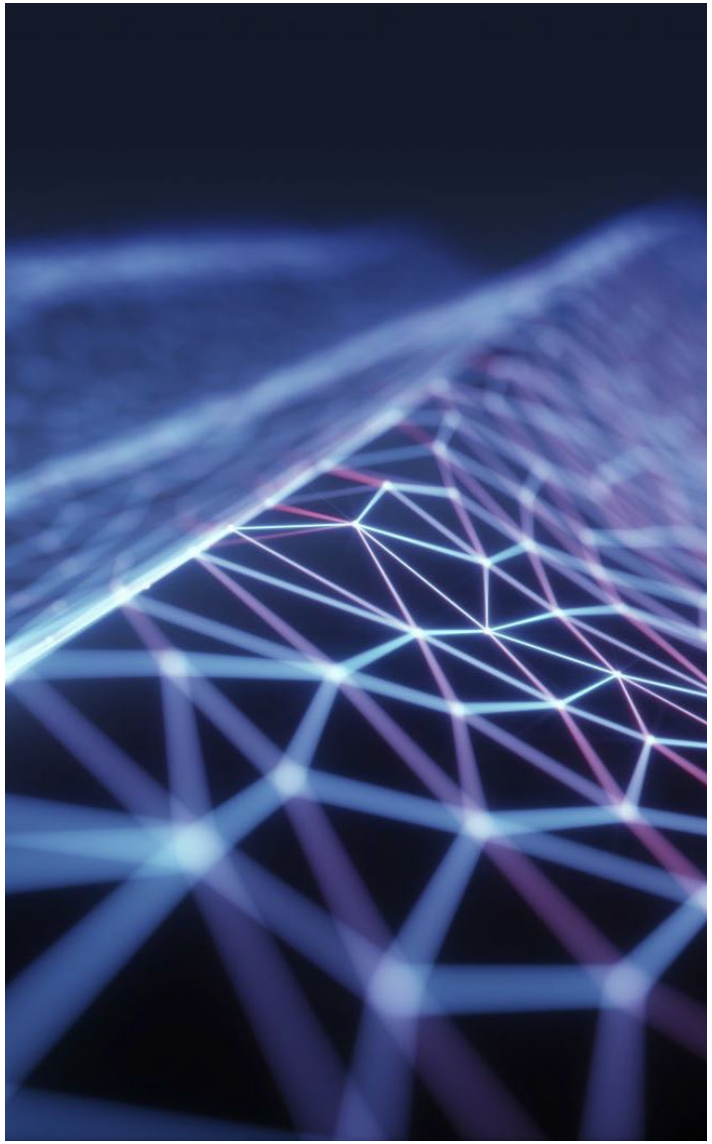
Gender as Predictive of Latent Class Membership

We have gender as a covariate

- We are predicting whether females (males) are more likely to be in class
  - 1 v 2
  - 1 v 3
  - 2 v 3
- But, the software selects only **one class** for comparison (class 1)

What do you expect

- Recall we have three classes…



Logistic regression refresher

- Logistic multinomial regression coefficient equals -0.3406
  - Then log odds ratio of being in class 2 (compared with reference class 1) is -0.3406 smaller for females compared to males
  - So, odds ratio is smaller by a factor exp(-0.3406) = 0.71
  - The % change in the odds is [exp(-.3406) – 1] * 100
  - So, odds are 30% smaller for females

Add ‘gender’ to predict membership in Greek Political Participation classes

- Female is 0/Male is 1
- The odds of being in class 1 versus class 3 are 1.93 times higher for males but the result is not statistically significant (*p* = .099)
- The odds of being in class 2 versus class 3 are 1.37 times higher for males but the result is not statistically significant (*p* = .082)
- Recall class 3 is the class associated with low levels of participation across all variables
- → males are more likely to participate in politics in Greek elections (not surprising)

Logistic Regression Results

	Odds ratio	Coefficient	*p*
Class1/3.(Intercept)	0.01	-4.21	< .001
Class1/3.Gender	1.93	0.66	0.099
Class2/3.(Intercept)	0.12	-2.16	< .001
Class2/3.Gender	1.37	0.31	0.082

DOING EFA

- The null hypothesis is that the model derived covariance matrix is the same as the observed covariance matrix
  - Do not want to reject the null, you want these to be equivalent!
  - The implied relationships are close to the observed relationships
  - I.e.. Does the model produce an estimated covariance matrix that is close to our sample covariance matrix
  - **RQ**: To what extent is there consistency between our estimated population covariance matrix produced by the model and the observed covariance matrix of our sample
- The idea is to estimate the parameters so that the discrepancy between the sample covariance matrix and the implied covariance matrix is minimal

ESTIMATING SEM

- We have an implied covariance matrix and an observed covariance matrix, the bigger the discrepancy the more likely the model is not fitting the data

		X1	X2	X3	X4
Empirical	X1	4.667	1.542	2.458	1.375
	X2	1.542	3.523	1.965	.583
	X3	2.458	1.965	3.657	1.375
	X4	1.375	.583	1.375	2.583
Implied	X1	4.480	1.493	2.391	1.068
	X2	1.493	3.382	1.849	.826
	X3	2.391	1.849	3.510	1.323
	X4	1.068	.826	1.323	2.480
Discrepancy	X1	.187	.049	.067	.307
	X2	.049	.141	.116	-.243
	X3	.067	.116	.147	.052
	X4	.307	-.243	.052	.103

PARAMETER ESTIMATE EVALUATION

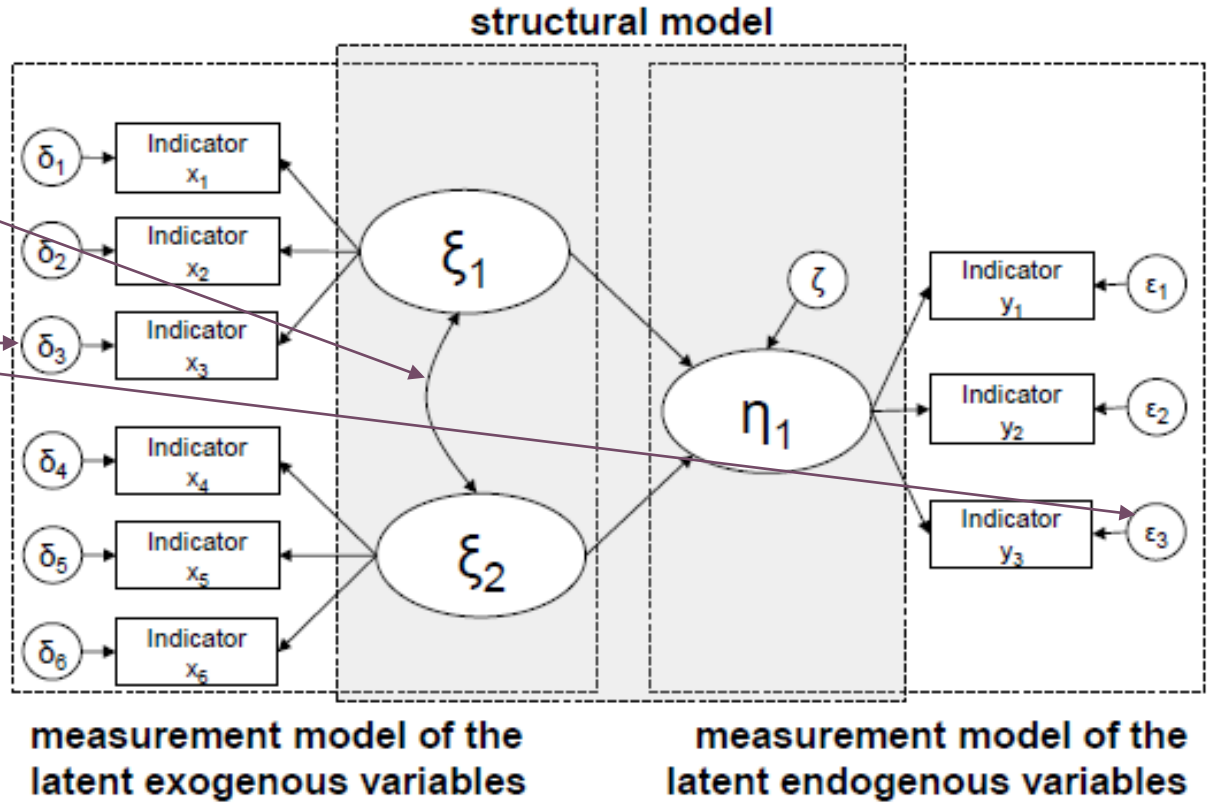
- Standardized factor loadings, when the observed variable loads onto only one latent construct, are interpreted as correlations and represent the relationship between the observed and latent variables.
  - When standardized factor loadings in this situation are squared, they are interpreted as the proportion of variance of the latent variable that is explained by the observed variable.
  - A standardized factor loading of at least .7 is recommended as, when squared, this is interpreted as the latent variable is explaining about 50% of the variance of the observed variable

THE STRUCTURAL AND MEASUREMENT MODEL

Covariance is part of measurement model

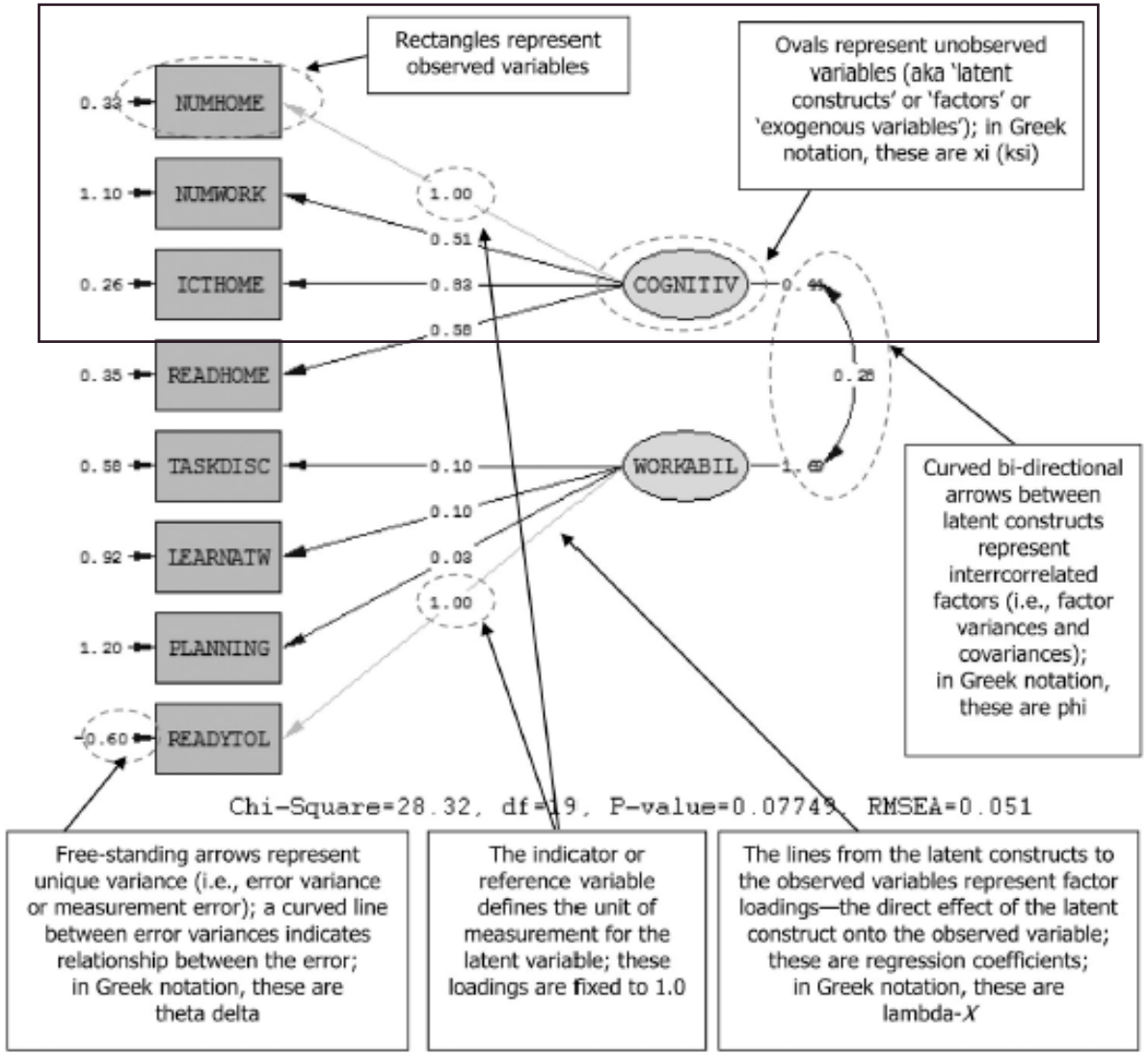
Unobserved error

Start with measurement model and assess fit before moving towards the structural model



CFA MODEL FROM A BOOK

- Each observed variable loads on only one factor
- All cross loadings are set to 0
  - EFA v CFA
  - We don't expect a high relationship going across indicators within factors
  - The paths are constrained or fixed to 0
- Rectangles are observed variables
- Ovals (including error) are latent



CFA MODEL

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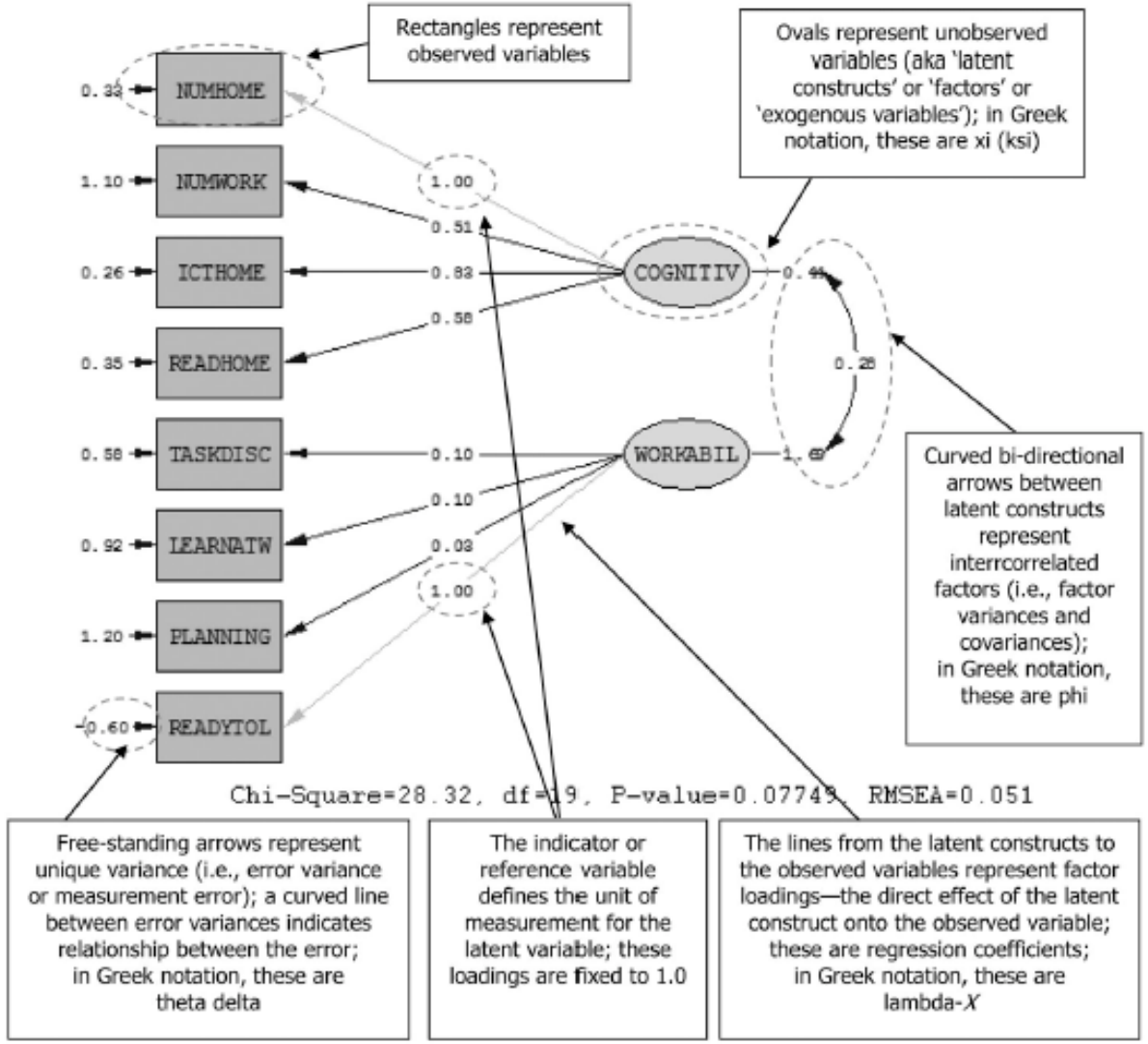
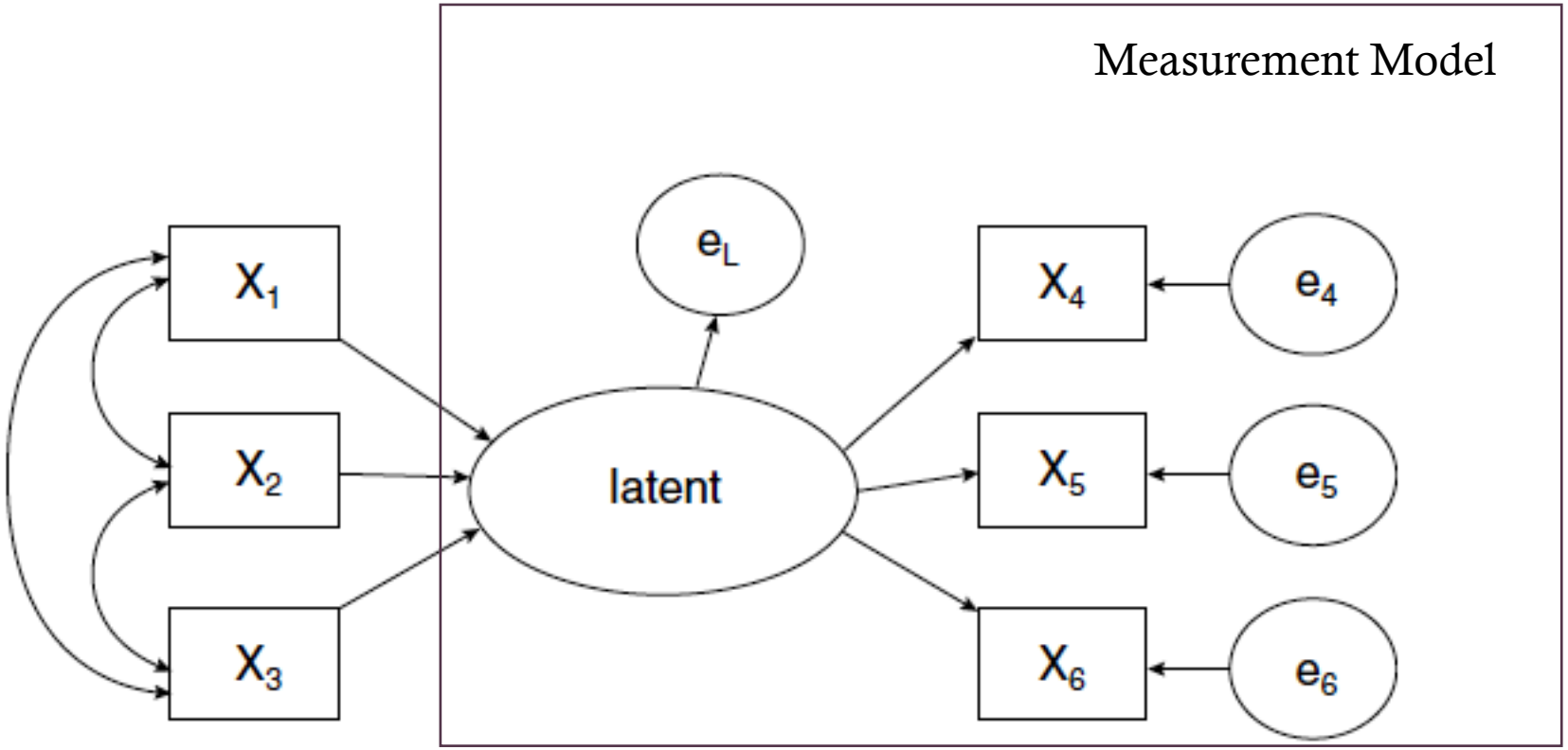
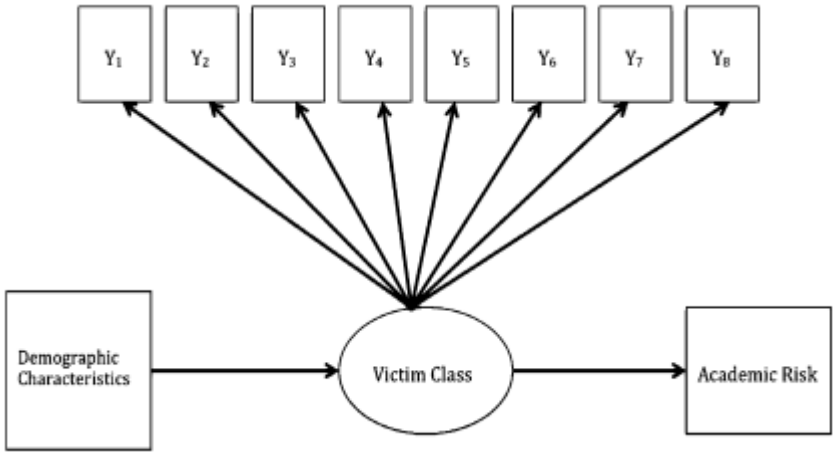


MULTIPLE INDICATOR–MULTIPLE CAUSE (MIMIC) MODEL



LATENT CLASS MIXTURE MODELS

- RQ: LCA is good to construct typologies
- How do families experience ACEs?
- How do children experience F2F and cyber-bullying?



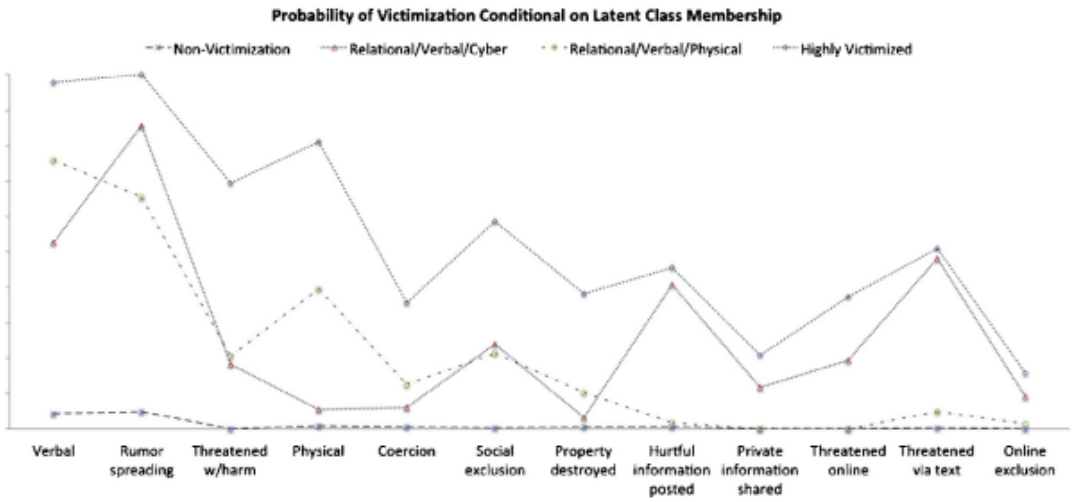


Fig. 2. Conditional response probabilities.

GROWTH MIXTURE MODELING

- RQ: How does depression manifest developmentally?
- Example: Do all people follow the same trajectory?

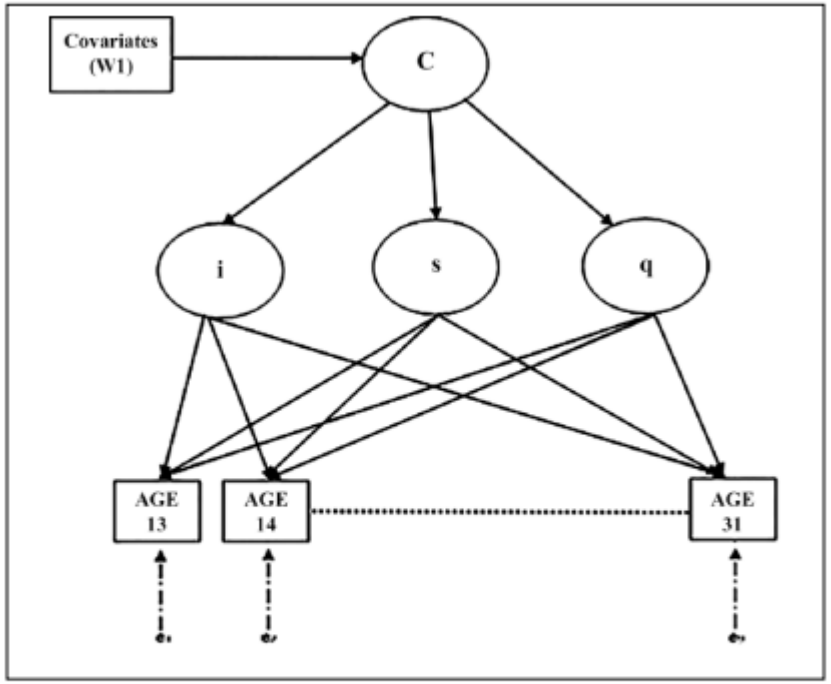


Figure 1. Growth mixture model of CES-9 depressive symptom trajectories among maltreated children.
Note. Model includes proximal Wave 1 variables (e.g., gender, delinquency) as predictors of class membership. CES = Center for Epidemiological Studies.

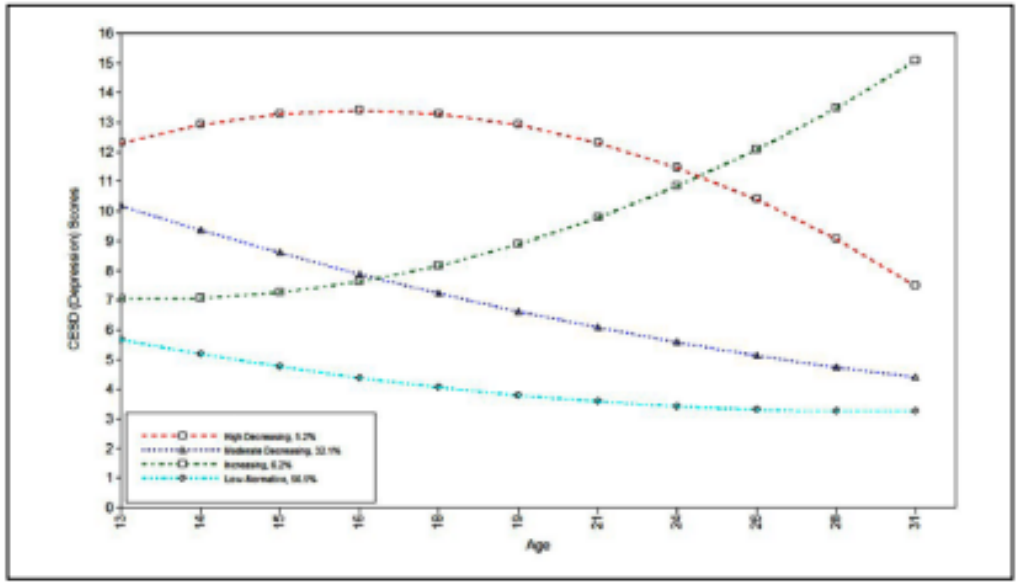
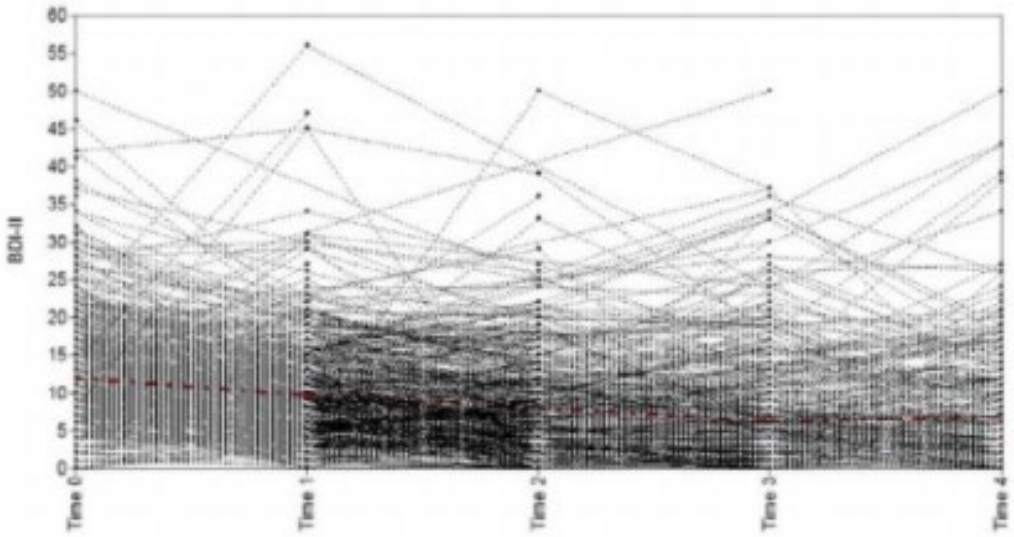
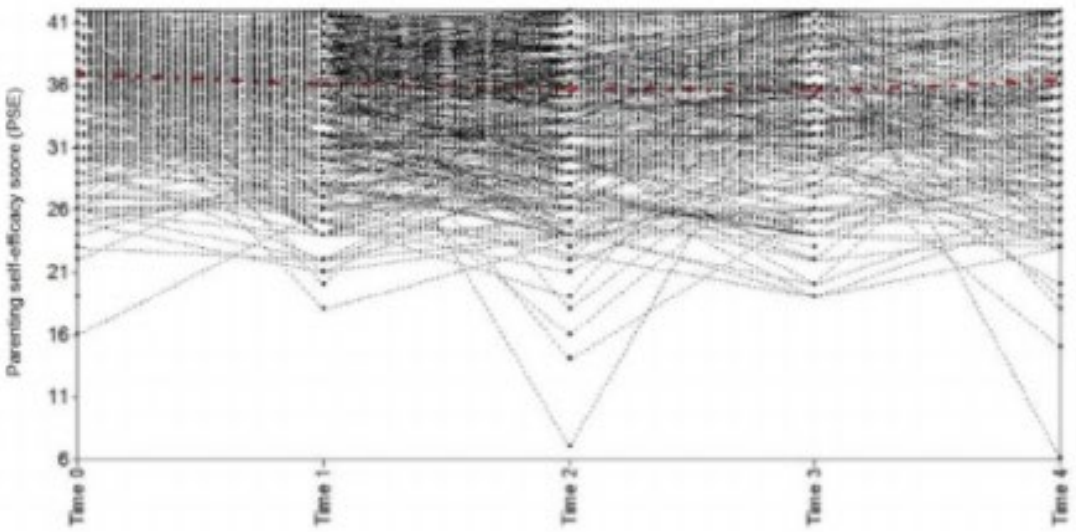


Figure 2. Mean growth curves for the four trajectory classes.
Note. Legend shows percentage of individuals comprising class. CES-D = Center for Epidemiological Studies of Depression Scale.

GROWTH MIXTURE MODELING

- Goals were to explore
  - Potential heterogeneity in **dual trajectories** of PSE and depressive symptoms among first-time postpartum mothers and
  - Whether the dual trajectory classes are differentially associated with poor outcomes in early childhood

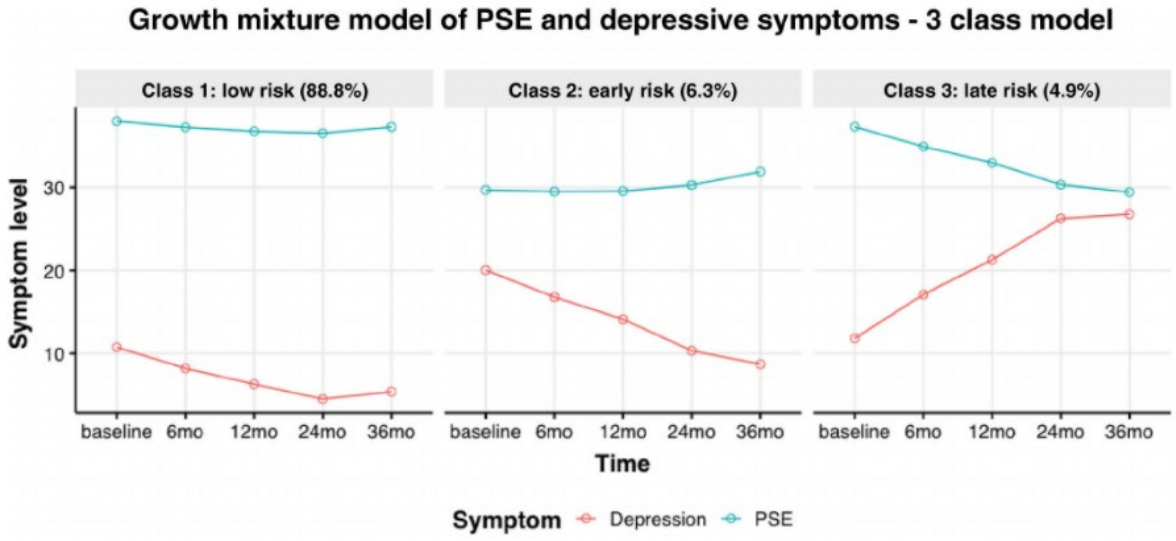


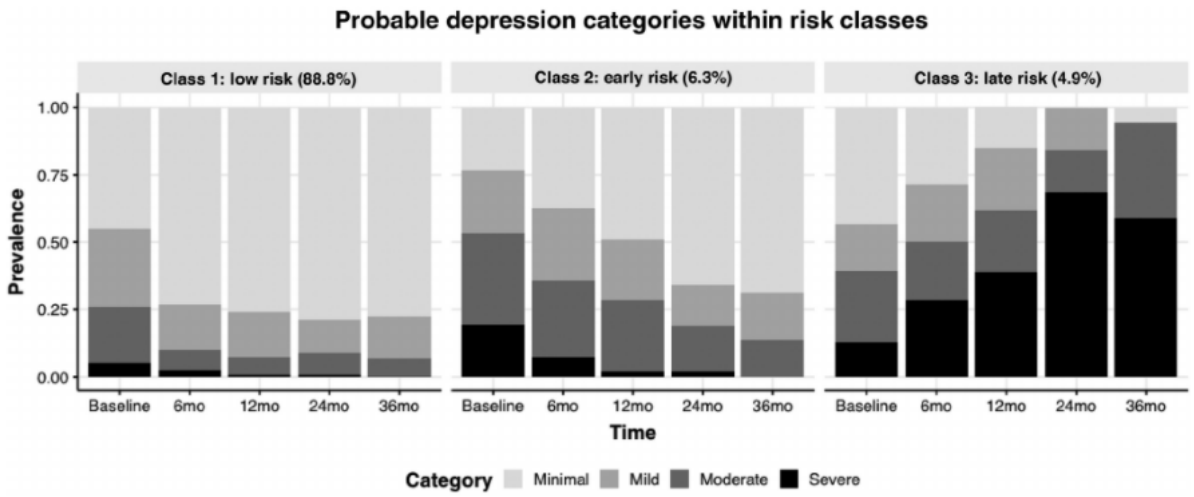


Barboza, G. E., & Schiamberg, L. (2021). Dual trajectories of parenting self-efficacy and depressive symptoms in new, postpartum mothers and socioemotional adjustment in early childhood: A growth mixture model. Infant mental health journal, 42(5), 636-654.

GROWTH MIXTURE MODELING

- Three classes of dual trajectories found
- Predictive validity established as low-risk classes had very low levels of moderate to severe depression compared to the high-risk classes





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GROWTH MIXTURE MODELING

- Subgroup analysis showed that children of mothers in the low-risk class showed significantly fewer total externalizing problem behaviors compared with the higher risk classes

TABLE 6 Mean (SD) scores of child behavior problems (ITSEA) by dual PSE and depressive symptom trajectory groups

CBCL	Class 1: low risk[a] ($n = 557$)	Class 2: early risk[b] ($n = 89$)	Class 3: late risk[c] ($n = 34$)	Group comparisons
Externalizing Domain Total	52.276 (0.972)	62.407 (2.91)	64.110 (4.707)	c > a*, b > a*
Internalizing Domain Total	51.451 (0.899)	55.716 (2.63)	57.411 (4.09)	ns. diff.
Aggression/Defiance cut score	0.186 (0.053)	0.435 (0.131)	0.621 (0.171)	b > a*, c > a*
Depression/Withdrawal cut score	0.225 (0.044)	0.507 (0.154)	0.635 (0.123)	a < b*, a < c*

sd = standard deviation. All child behavior problems were measured at 36 months (i.e., age 3 years old).

[a]Low-risk class.

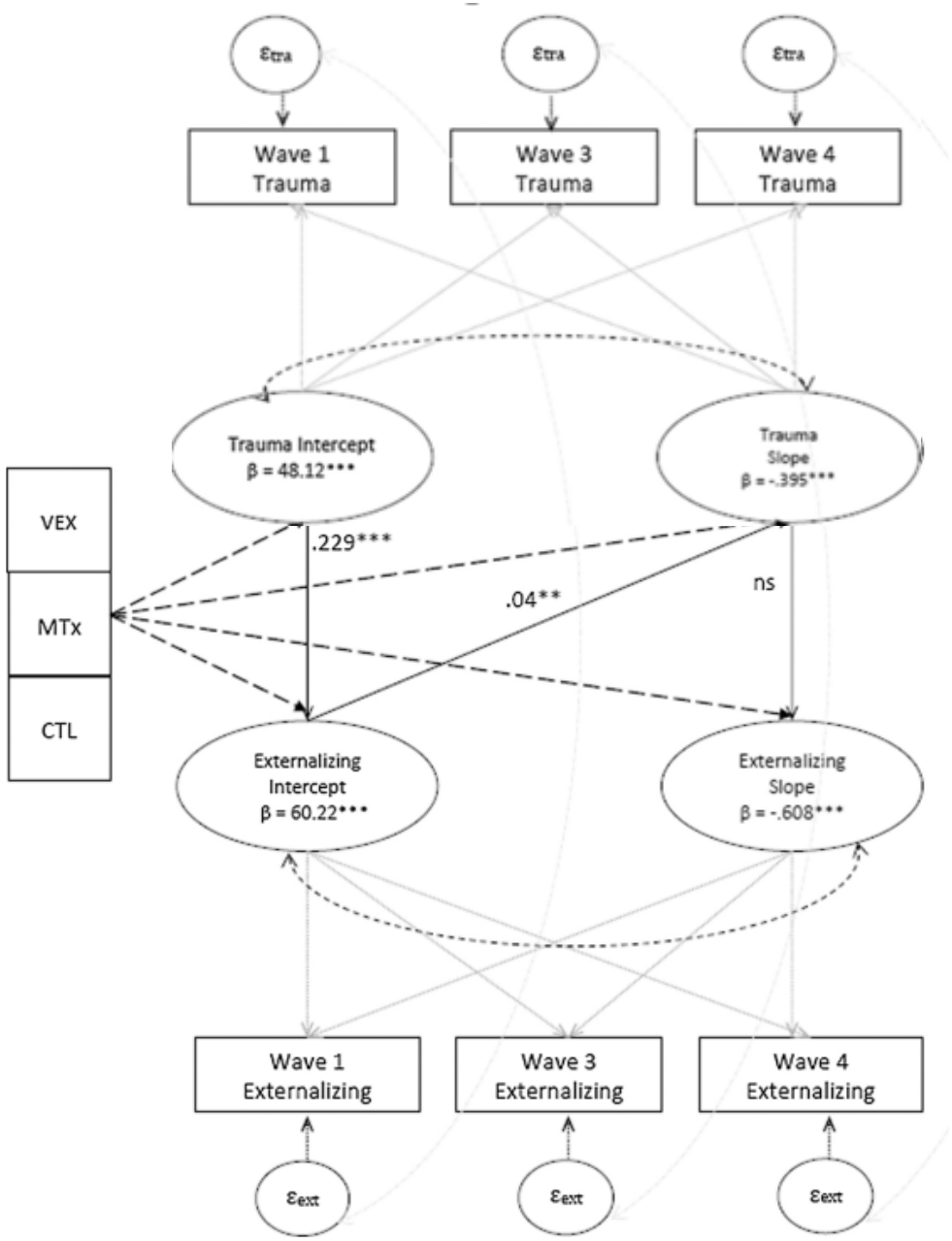
[b]Early-risk class.

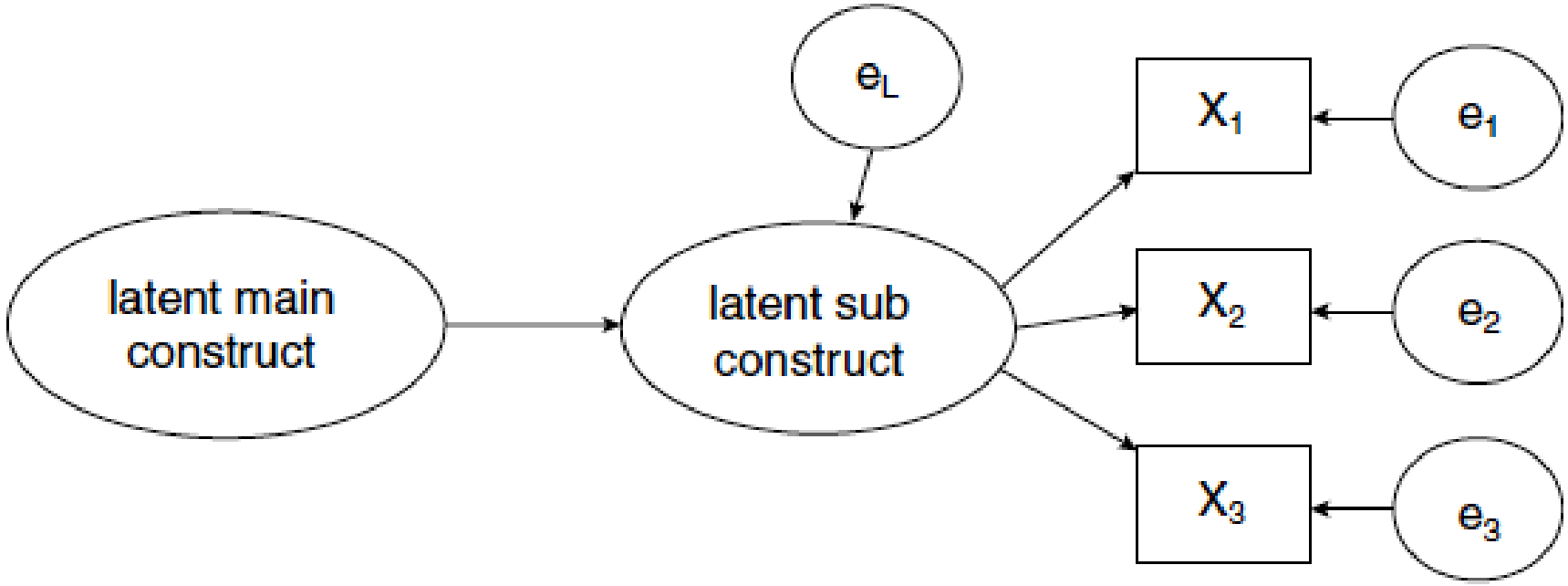
DYNAMIC FACTOR MODELS

- Include repeated measures completed on two or more occasions
- Allow change in the latent factor to be assessed over time

LATENT GROWTH MODELS

- Example of latent growth curve model of PTS
- Example parallel process model that explores how growth in one symptom affects growth in another





SECOND ORDER CFA

The first order latent factor explains a higher order latent factor

SECOND ORDER GROWTH CURVE MODEL

- Here is a growth curve model with intercept, slope and quadratic
- The growth is believed to mediate the relationship between 'readiness to parent' and child behavior
- PS is comprised of three subscales
- Perceived SS moderates the relationship between readiness to parent and growth in PS

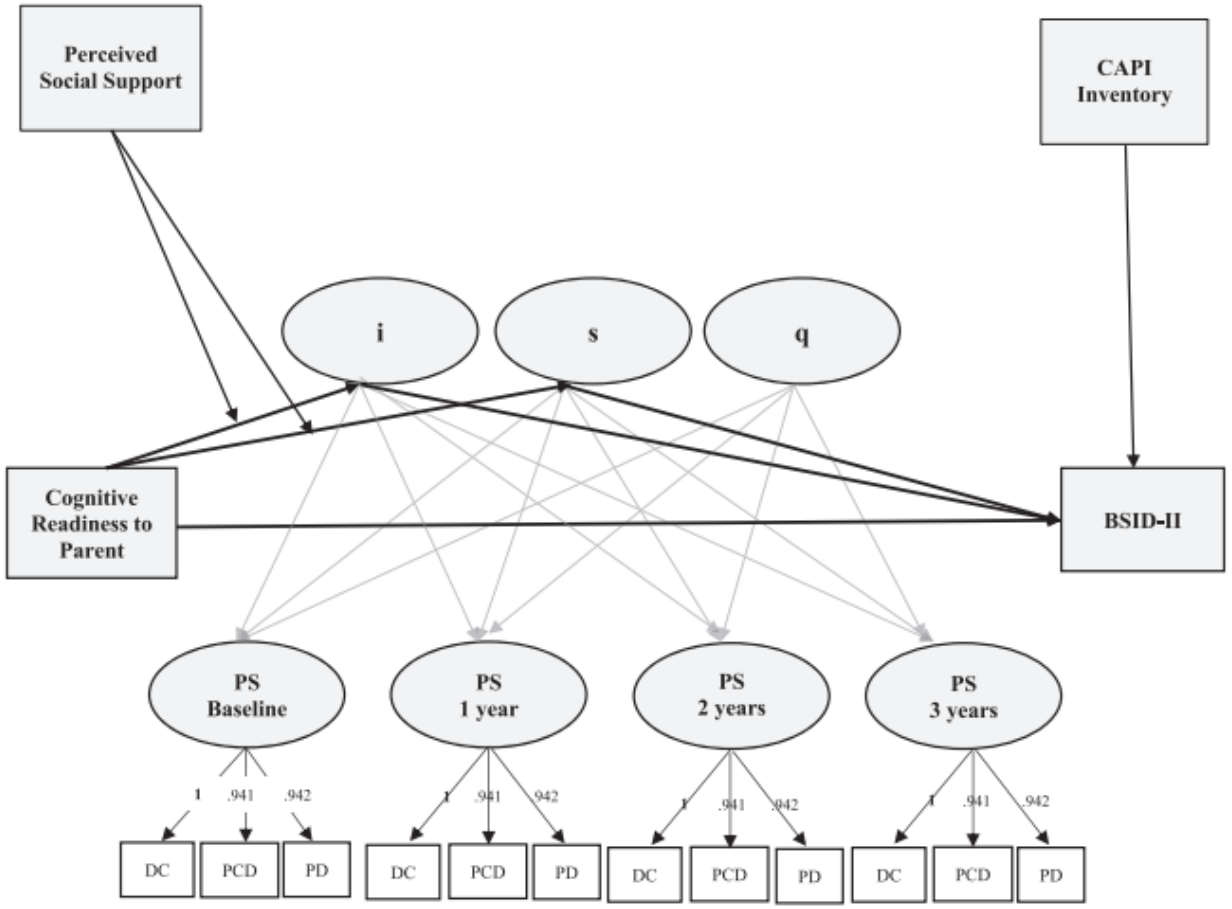


Fig. 1. Structural equation model including a second order linear growth curve for parenting stress (PS) symptoms. All coefficients are unstandardized parameter estimates based on maximum likelihood estimation. parenting stress = Parenting Stress; DC = Difficult Child; PCD = Parent-Child Dysfunction; PD = Parental Distress; i = Intercept; s = Slope; q = Quadratic. Autocorrelation between parenting stress factors and growth factor covariances were included in the model. Partial strong variance was specified under the marker variable approach. Bolded values indicate fixed parameters. Additional variables included in the model are social support, difficult life circumstances, maternal age, marital status, race/ethnicity, receipt of welfare benefits and education. Three measures of cognitive readiness to parent are hypothesized to be positively related to initial levels of parenting stress (intercept) and change (increases) in parenting stress (slope) over time. In turn, parenting stress growth factors (intercept and slope) are hypothesized to predict lower levels of social-emotional competence among children at age 3. In this model, parenting stress mediates the relation between cognitive readiness to parent and future social-emotional competences in children.

FA IN JASP

- I have an annotated file that demonstrates how to conduct and write-up EFA
- I have an annotated file that demonstrates how to conduct and write-up LCA
- Any other models we should cover?
- Next up: Sampling